RETROENGINE BETA ENCYCLOPEDIA

Volume 2: Editor Suite & Viewport Workspace Guide

1. DOCKSPACE TOPOLOGY & NDC COORDINATE MAPPING

RetroEngine provides a multi-panel docking layout system built upon a binary split tree (DockNode). Every panel beneath the top menu bar is represented as a docked tile. The master layout centers around a unified **Center Tile** that tab-stacks the 3D Viewport (DockPanel.SCENE) and the Animator State Machine Canvas (DockPanel.ANIMATOR). When the user switches tabs between SCENE and ANIMATOR in the Center Tile, the left and right docking columns automatically swap their companion panels.

DockPanel Identifier	Panel Region	Companion Swap Behavior
`SCENE`	**Center Tile (VIEWPORT)**	Tabbed with ANIMATOR. Active when inspecting 3D scene world and entities.
`ANIMATOR`	**Center Tile (ANIMATOR)**	Tabbed with SCENE. Active when editing skeletal animation state machines.
`HIERARCHY`	**Left Column (SCENE_HIERARCHY)**	Active when SCENE tab is selected. Displays scene entity tree.
`INSPECTOR`	**Right Column (INSPECTOR)**	Active when SCENE tab is selected. Displays entity component reflection.
`PARAMETERS`	**Left Column (ANIMATOR)**	Swaps into Left Column when ANIMATOR tab is active. Edits float/int/bool/trigger parameters.
`ANIMATOR_INSPECTOR`	**Right Column (ANIMATOR)**	Swaps into Right Column when ANIMATOR tab is active. Details state transitions & dope sheet.
`ENGINE_CONSOLE`	**Bottom Panel (CONSOLE)**	Docked at bottom of screen. Logs engine stdout/stderr and performance telemetry.
`ASSET_BROWSER`	**Bottom Panel (ASSET_BROWSER)**	Docked at bottom of screen. Displays disk-mirrored Assets/ directory tree.

> **Center Tile Tabbed Workspace Rule**
>
> The 3D Scene Viewport and Animator Graph Canvas occupy the exact same screen bounds in the center of the EditorWorkspace. Activating the ANIMATOR tab automatically swaps the left panel to PARAMETERS and the right panel to ANIMATOR_INSPECTOR. Re-activating the SCENE tab restores HIERARCHY on the left and INSPECTOR on the right.

2. MULTI-REGION INPUT ROUTER & SHORTCUTS

Input dispatching is orchestrated by EditorInputRouter, establishing a single-authority event priority pipeline. Input events flow hierarchically through Modal Overlays, Context Menus, Active Dock Nodes, and Panel Delegates. Keyboard chords are managed by HotkeyRouter and formatted for the host platform via PlatformShortcuts.

Action Command	macOS Chord	Windows / Linux Chord	Internal Handler & Dispatch Rule
Save Scene	⌘S	`Ctrl + S`	Prompts native dialog to serialize scene binary (.rscene).
Open Scene	⌘O	`Ctrl + O`	Prompts native dialog to load scene binary (.rscene).
Undo Action	⌘Z	`Ctrl + Z`	Pops and executes top command from CommandStack.
Redo Action	⌘⇧Z	`Ctrl + Shift + Z`	Re-executes previously undone command from CommandStack.
Duplicate Entity	⌘D	`Ctrl + D`	Instantiates deep copy of selected Entity via EntitySpawnCommand.
Delete Entity	⌫ (Delete)	`Del / Backspace`	Deletes selected Entity via EntityDeleteCommand (ignored if text field focused).
Toggle Console	~ (Tilde)	~ (Tilde)	Toggles visibility of EngineConsolePanel overlay.
Frame Selected	F	F	Centers ViewportCamera onto selected entity bounds.
Reset Viewport	⇧F	Shift + F	Resets ViewportCamera to default startup pose.
Gizmo Mode: Move	1 or W	1 or W	Sets gizmo mode to TRANSLATE (disabled during WASDQE flight).
Gizmo Mode: Rotate	2 or E	2 or E	Sets gizmo mode to ROTATE (disabled during WASDQE flight).
Gizmo Mode: Scale	3 or R	3 or R	Sets gizmo mode to SCALE (disabled during WASDQE flight).
Toggle Snapping	Cmd (Hold)	Ctrl (Hold)	Inverts default snapping. Snapping is ON by default; holding Cmd/Ctrl disables it.

Focus Lock & Text Field Protection

When an Inspector text field is actively focused (InspectorPanel.isTextFieldFocused() == true), shortcut routing in HotkeyRouter is completely locked. Keypresses like 'W', 'S', or 'Delete' edit the string buffer rather than triggering viewport flight or entity deletion.

3. HIERARCHY PANEL & SCENE GRAPH MANAGEMENT

The HierarchyPanel displays a flattened, zero-allocation tree view of all active entities in the SceneWorld. It supports parent-child relationships, entity selection, drag-and-drop re-parenting, and context menu spawning.

Hierarchy Interaction	User Input / Mouse Action	Engine System & Validation
Select Entity	`Left-Click Entity Row`	Updates SceneWorld.setSelectedEntity() and highlights row.
Multi-Select	`Ctrl / Cmd + Click Row`	Appends entity to active multi-selection list.
Re-parent Entity	`Drag Row onto Target Row`	Validates isCycleSafeParent() before dispatching ParentChangeCommand.
Re-order Entity	`Drag Row between Rows`	Inserts entity before/after target in sibling list.
Context Menu	`Right-Click Panel / Entity`	Opens ContextMenuOverlay (Spawn Cube, Light, Camera, Delete, Rename).
Rename Entity	`Double-Click Name / F2`	Opens inline text edit box; commits via EntityRenameCommand.

> **Cycle-Safe Re-parenting Guarantee**
> Before applying a re-parenting operation, HierarchyPanel executes isCycleSafeParent(candidateParent, entity). If the candidate parent is a descendant of the dragged entity, the operation is rejected, preventing cyclic scene graph loops.

4. INSPECTOR PANEL & @INSPECTORPROPERTY REFLECTION

The InspectorPanel dynamically renders component fields for the selected entity using reflection and metadata from ComponentSchema. Fields annotated with @InspectorProperty are rendered with dedicated UI controls.

Property Control Type	Supported Field Types	UI Interaction & Command Generation
Numeric Stepper	`float, int, double`	Click +/- or drag label to adjust value. Dispatches ModifyPropertyCommand.
Toggle Switch	`boolean`	Click pill switch to toggle true/false. Dispatches ModifyPropertyCommand.
Vector3 Control	`float[3], Vector3f`	Renders X (Red), Y (Green), Z (Blue) numeric inputs.
Color Picker	`Color, float[4]`	Displays color swatch box; clicking opens RGBA color picker overlay.
Asset Reference	`Texture, Mesh, AudioClip`	Displays asset name/UUID; drag asset from Asset Browser to assign.
Enum Dropdown	`Enum constants`	Click dropdown button to select enum value.

Transactional Undo/Redo for Inspector Edits

Modifying any field in the Inspector generates a ModifyPropertyCommand containing the old value, new value, and reflection setter. This ensures that every property tweak can be instantly reverted via Cmd + Z / Ctrl + Z.

5. 3D VIEWPORT NAVIGATION & CAMERA CONTROLS

The ViewportPanel hosts the primary 3D rendering canvas. Viewport navigation is driven by ViewportCamera, offering Unity-style 6-DOF WASDQE flight, orbit, pan, and framing.

Navigation Mode	Mouse & Key Controls	Camera Pacing & Behavior
6-DOF WASDQE Flight	`Hold RMB or MMB + WASDQE`	W/S: Forward/Back, A/D: Strafe Left/Right, Q/E: Down/Up. Speed controlled by mouse wheel.
Camera Orbit	`Hold RMB + Drag / Alt + LMB Drag`	Orbits camera around current pivot point (adjusts pitch & yaw).
Camera Pan	`Hold MMB + Drag / Shift + RMB Drag`	Pans camera along local X/Y viewport axes.
Zoom	`Mouse Scroll Wheel`	Dollys camera forward/backward along view vector.
Frame Selected (F)	`Press F`	Glides camera pivot to center on selected entity's bounding box.
Reset Camera (Shift+F)	`Press Shift + F`	Resets camera to default startup position and orientation.

Picture-in-Picture (PiP) Camera Preview

When an entity containing a Camera component is selected, a 640x360 Picture-in-Picture (PiP) inset preview is automatically rendered in the lower-right corner of the Viewport, allowing real-time framing checks.

6. 3D TRANSFORM GIZMOS & SNAPPING ENGINEERING

Interactive 3D transform manipulation is handled by GizmoInteractor, GizmoMath, and GizmoRenderer. Gizmos support Translate, Rotate, and Scale modes across X (Red), Y (Green), and Z (Blue) axes.

Gizmo Mode	Shortcut	Default Snap Increment	Snapping Inversion Rule
TRANSLATE	`1 or W`	`0.5 Units`	Snapping is ON by default. Holding Cmd/Ctrl disables snapping for free movement.
ROTATE	`2 or E`	`15.0 Degrees`	Snapping is ON by default. Holding Cmd/Ctrl disables snapping for free movement.
SCALE	`3 or R`	`0.25 Scale`	Snapping is ON by default. Holding Cmd/Ctrl disables snapping for free movement.

Bottom-Center Mesh Bounds Alignment

Translate gizmos automatically anchor to the bottom-center of an entity's mesh bounding box rather than its geometric centroid. This makes placing objects flush onto terrain or floors intuitive and precise.

7. PIXEL-EXACT COLOR-ID PICKING & DEBUG RENDERER

RetroEngine uses an offscreen 32-bit GPU Color-ID picking pass for 3D entity selection. Additionally, ColliderDebugRenderer draws immediate-mode 3D wireframes in viewport space.

Wireframe Category	Active / Unselected Color	Selected / Highlighted Color	Trigger Color
BoxCollider	`Green (R:0.20, G:0.90, B:0.50)`	`Cyan (R:0.35, G:0.75, B:1.00)`	`Yellow (R:0.95, G:0.85, B:0.20)`
SphereCollider	`Green (R:0.20, G:0.90, B:0.50)`	`Cyan (R:0.35, G:0.75, B:1.00)`	`Yellow (R:0.95, G:0.85, B:0.20)`
CapsuleCollider	`Green (R:0.20, G:0.90, B:0.50)`	`Cyan (R:0.35, G:0.75, B:1.00)`	`Yellow (R:0.95, G:0.85, B:0.20)`
CharacterController	`Green (Grounded)`	`Cyan (Selected)`	`Red/Orange (Airborne)`
VehicleSuspension	`Green (Grounded Ray)`	`Cyan (Selected)`	`Red (Airborne Ray)`

Wireframe Color Convention Standard

All active non-trigger colliders render as Green wireframes in the Viewport. When selected, they turn Cyan. All trigger volumes (isTrigger == true) render as bright Yellow wireframes regardless of selection state.

8. ANIMATOR GRAPH PANEL & STATE MACHINE CANVAS

When the **ANIMATOR** tab is selected in the Center Tile, the center view renders the AnimatorGraphPanel. This panel provides a node-based state machine editor for configuring animation clips, transitions, and blend trees.

Node / Connection Type	Canvas Representation	Functionality & Rules
ENTRY Node	`Green Card`	Default entry point when state machine starts. Connects to initial state.
ANY STATE Node	`Teal Card`	Global state. Transitions from ANY STATE fire regardless of active state.
AnimatorState Node	`Slate Card`	Holds animation clip reference, playback speed, and loop settings.
AnimatorTransition Link	`Bezier Arrow`	Connects source to target state. Contains transition conditions and crossfade duration.

Companion Panel Swapping

Selecting the ANIMATOR tab automatically swaps the left panel to PARAMETERS (edits float, int, bool, trigger parameters) and the right panel to ANIMATOR_INSPECTOR (edits transition conditions and dope sheet keyframes).

9. ANIMATOR INSPECTOR & DOPE SHEET TIMELINE

The AnimatorInspectorDelegate renders the right-column companion panel when the Animator workspace is active. It provides transition condition management and a Dope Sheet timeline for recording keyframes.

Dope Sheet Feature	Control / Keybinding	Description & Usage
Keyframe Pose (K)	`Press K`	Keys selected bone transform into active clip at playhead position.
Step Frame Back (,)	`Press , (Comma)`	Steps timeline playhead 1 frame backward.
Step Frame Forward (.)	`Press . (Period)`	Steps timeline playhead 1 frame forward.
Scrub Playhead	`Click/Drag Timeline Ruler`	Evaluates pose at time t and updates skeletal pose in Viewport.
Export Clip (.ranim)	`Click Export Button`	Serializes keyframe tracks to binary .ranim asset format.

> **Zero-Allocation Bone Palette Skinning**
>
> Recorded keyframe tracks are baked into 64-matrix bone palettes. At runtime, AnimationSystem uploads palettes directly to GPU uniform buffers, executing zero-allocation hardware vertex skinning.

10. AUDIO ATTENUATION & EDITOR FIELD GUIDE

Spatial 3D audio is driven by AudioSource components, supporting distance attenuation and panning. This section concludes Volume 2 with an Editor Troubleshooting Field Guide.

Troubleshooting Area	Observed Issue / Anomaly	Recommended Resolution
Layout Corruption	`Panels overlap or disappear`	Press ~ to open console, then run Reset Dock Layout.
Gizmo Not Responding	`Gizmo handles unclickable`	Ensure camera flight mode (RMB/MMB) is released.
Shortcuts Locked	`Keys W/S/Delete fail to fire`	Check if an Inspector text field is currently focused; press Enter or Esc.
Selection Mismatch	`Clicking entity selects wrong object`	Check for overlapping colliders or multi-mesh bounds.
Audio Muted	`3D audio source silent`	Verify minDistance/maxDistance bounds and AudioListener position.

Summary & Next Volume

This concludes Volume 2 of the RetroEngine Beta Encyclopedia. Refer to Volume 3 (Graphics Hardware Interface & Pipeline Engineering) for Vulkan 1.3/Metal 3 details, and Volume 4 (Beta QA Testing & Field Guide) for comprehensive testing protocols.